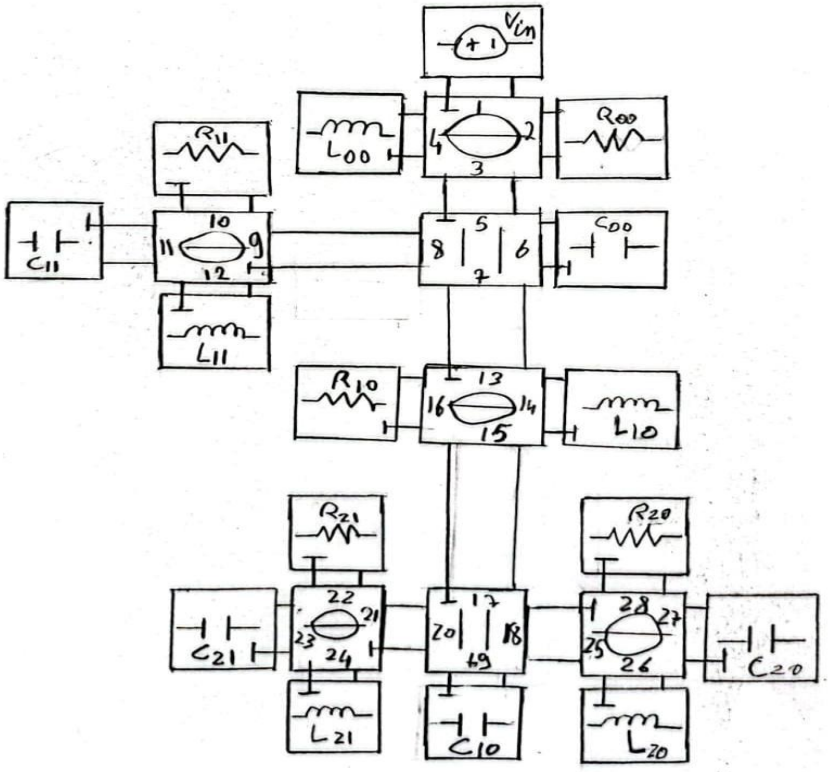
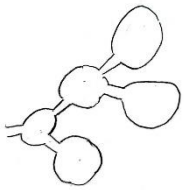
SASP (SSSP) - Homework 2

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1.0 WDF Scheme and Free Parameters



helmholtz imagination



* Adaptors:

* peripheral Branches:

- $Z_{25} = Z_{26} + Z_{27} + Z_{28}$
- $Z_{21} = Z_{22} + Z_{23} + Z_{24}$
- $Z_9 = Z_{10} + Z_{11} + Z_{12}$

* first parallel adaptor:

- $Z_{17} = (Z_{18}^{-1} + Z_{19}^{-1} + Z_{20}^{-1})^{-1}$
- $Z_{18} = Z_{25}$, $Z_{20} = Z_{21}$

* Branch 10 Series adaptor:

- $Z_{15} = Z_{17}$
- $Z_{13} = Z_{14} + Z_{15} + Z_{16}$

* Second parallel adaptor:

- $Z_7 = Z_{13}$, $Z_8 = Z_9$
- $Z_5 = (Z_6^{-1} + Z_7^{-1} + Z_8^{-1})^{-1}$

* the Root adaptor:

- $Z_1 = Z_2 + Z_3 + Z_4$

* Leaves:

* Resistors ($Z_R = R$):

- $Z_{28} = R_{20}$
- $Z_{22} = R_{21}$
- $Z_{16} = R_{10}$
- $Z_{10} = R_{11}$
- $Z_2 = R_{00}$

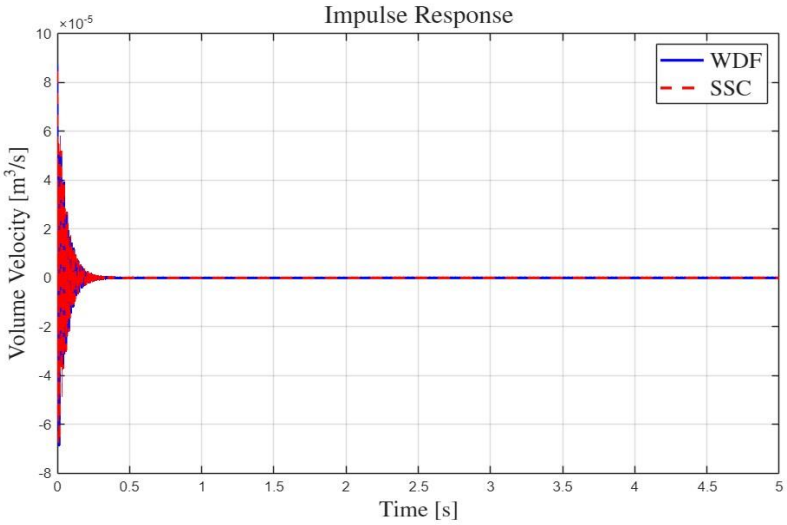
* inductors ($Z_L = \frac{2L}{T_s}$):

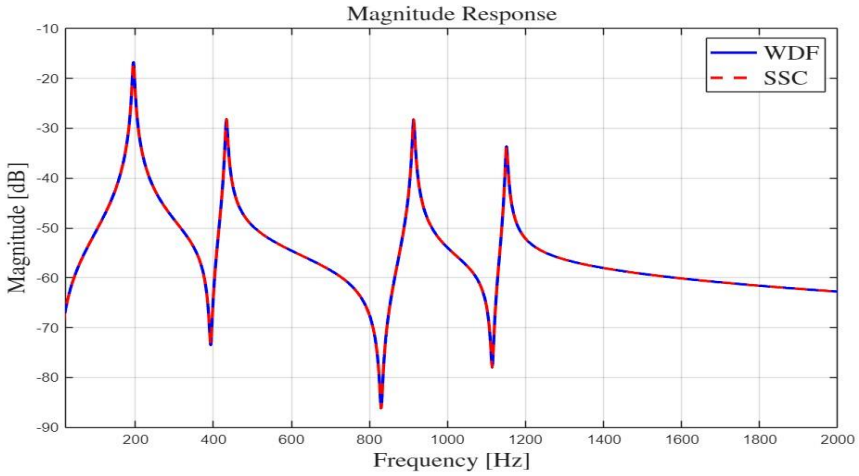
- $Z_{26} = \frac{2L_{20}}{T_s}$
- $Z_{24} = \frac{2L_{21}}{T_s}$
- $Z_{14} = \frac{2L_{10}}{T_s}$
- $Z_{12} = \frac{2L_{11}}{T_s}$
- $Z_4 = \frac{2L_{00}}{T_s}$

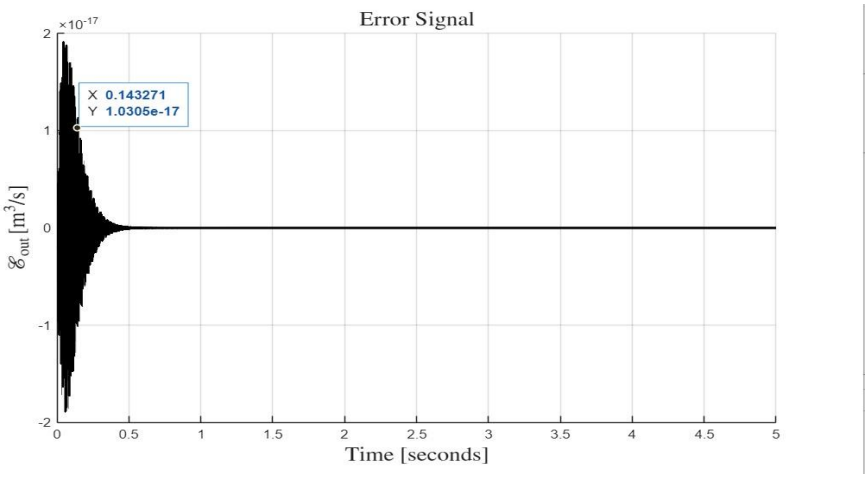
* capacitors ($Z_C = \frac{T_s}{2C}$):

- $Z_{27} = \frac{T_s}{2C_{20}}$
- $Z_{23} = \frac{T_s}{2C_{21}}$
- $Z_{19} = \frac{T_s}{2C_{10}}$
- $Z_{11} = \frac{T_s}{2C_{11}}$
- $Z_6 = \frac{T_s}{2C_{00}}$

2.0 Outputs and Clarifications:








```
NMSE = 2.48946e-25
```

The proposed Wave Digital Filter (WDF) scheme models the coupled Helmholtz resonator network using a fully explicit connection tree. To minimize computational complexity, the equivalent electrical components for the acoustic cavities and necks are grouped into standard 4-port Series and Parallel adaptors. I also tried the white noise signal and played with some parameters.

to calculate the scattering matrix for all ports at a junction simultaneously. I used the following two matrix functions:

For Series Adaptors:

$$\mathbf{S} = \mathbf{I} - \boldsymbol{\alpha}\mathbf{1}^T$$

For Parallel Adaptors:

$$\mathbf{S} = \mathbf{1}\boldsymbol{\alpha}^T - \mathbf{I}$$